

project

 Courses	 BAXZ2343 Computer Network
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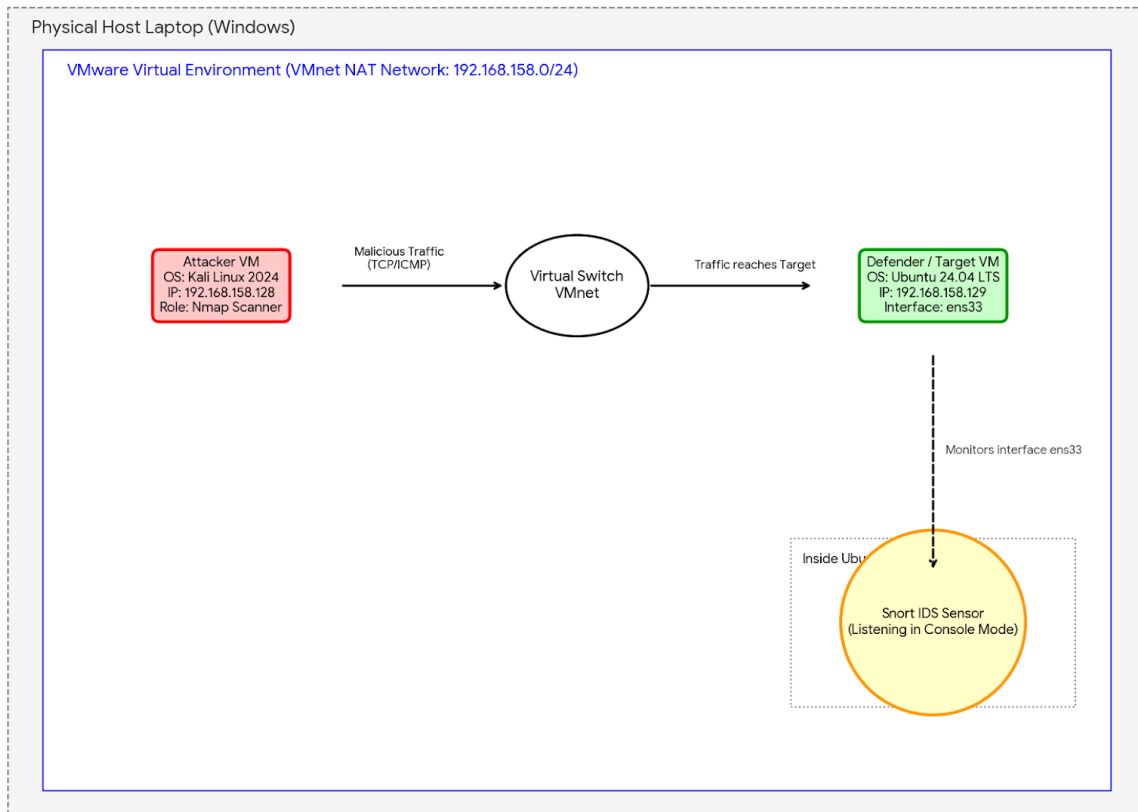
2.0 Methodology and Experimental Setup

2.1 System Architecture Overview

To simulate a controlled cyber-attack and intrusion detection scenario, a virtualized laboratory environment was designed. The methodology relies on a centralized NAT (Network Address Translation) configuration to allow isolated communication between virtual machines (VMs) while maintaining a secure separation from the external physical network.

2.2 Network Topology Diagram

The following diagram illustrates the logical connection between the Attacker (Kali Linux) and the Defender (Ubuntu), detailing the flow of malicious traffic through the virtual switch to the Snort IDS sensor.



2.3 Hardware and Software Specifications

The experimental setup consists of three primary layers: the Physical Host, the Virtual Network, and the Guest Operating Systems.

2.3.1 Physical Host Layer

- **Host Device:** Windows Laptop (Windows 11)
- **Hypervisor:** VMware Workstation Pro/Player
- **Role:** Provides the computing resources (CPU, RAM, Network Adapter) necessary to run the virtualized environment.

2.3.2 Virtual Network Layer

- **Network Type:** NAT (VMnet8)
- **Subnet:** 192.168.158.0/24
- **Gateway:** 192.168.158.2
- **Mechanism:** A virtual switch acts as the intermediary, ensuring that traffic generated by the Attacker VM is routed correctly to the Target VM without leaking onto the public internet.

2.3.3 Guest Systems (Virtual Machines)

Role	Hostname	OS Version	IP Address	Configuration Details
Attacker	Kali-VM	Kali Linux 2024	192.168.158.128	Equipped with penetration testing tools (Nmap, Ping) to generate traffic signatures.
Defender	Ubuntu-VM	Ubuntu 24.04 LTS	192.168.158.129	Hosts the Snort IDS application. The interface ens33 is configured in promiscuous mode.

2.4 Traffic Flow Logic

As depicted in Figure, the traffic flow for the experiment is designed as follows:

1. Generation: The Attacker VM initiates network scans (TCP/ICMP) targeting the Defender's IP.
2. Transmission: The traffic travels through the virtualized switch (VMnet).
3. Interception: The packet arrives at the Defender VM's network interface (ens33).
4. Monitoring: The Snort IDS sensor passively monitors ens33 to analyze the payload against specific rule sets (configured in Section 3.0).

Section 3.0: Implementation of Snort IDS

3.1 System Environment & Architecture

To simulate a real-world network attack scenario, a virtualized laboratory environment was created using VMware Workstation. The network consists of two distinct virtual machines (VMs):

- The Defender (Target): Ubuntu 24.04 LTS running Snort IDS.
- IP Address: 192.168.158.129
- Interface: ens33

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:98:4f:6d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.158.129/24 brd 192.168.158.255 scope global dynamic noprefixroute ens33
        valid_lft 1140sec preferred_lft 1140sec
    inet6 fe80::20c:29ff:fe98:4f6d/64 scope link
        valid_lft forever preferred_lft forever
```

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:98:4f:6d brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.158.129/24 brd 192.168.158.255 scope global dynamic noprefixroute ens33
        valid_lft 1140sec preferred_lft 1140sec
    inet6 fe80::20c:29ff:fe98:4f6d/64 scope link
        valid_lft forever preferred_lft forever
```

- The Attacker: Kali Linux
- IP Address: 192.168.158.128
- Tools Used: ping (Connectivity Test), nmap (Port Scanning).

```
(kali@kali)-[~/Desktop]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5c:8c:a6 brd ff:ff:ff:ff:ff:ff
    inet 192.168.158.128/24 brd 192.168.158.255 scope global dynamic noprefixroute eth0
        valid_lft 1598sec preferred_lft 1598sec
    inet6 fe80::105:f8e1:9241:d5db/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

```
(kali@kali)-[~/Desktop]
$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:5c:8c:a6 brd ff:ff:ff:ff:ff:ff
    inet 192.168.158.128/24 brd 192.168.158.255 scope global dynamic noprefixroute eth0
        valid_lft 1598sec preferred_lft 1598sec
    inet6 fe80::105:f8e1:9241:d5db/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

3.2 Installation Process

The installation was performed on the Ubuntu server using the default package repositories to ensure stability.

System Update & Installation The system repositories were updated, and the Snort package was installed using the following commands:

```
sudo apt update
```

```
sudo apt install snort -y
```

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/Desktop$ sudo apt update
[sudo] password for ubuntu:
Hit:1 http://my.archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://my.archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:3 http://my.archive.ubuntu.com/ubuntu noble-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
79 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ubuntu-VMware-Virtual-Platform:~/Desktop$ sudo apt install snort -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  libdaq2t64 libdumbnet1 libluajit-5.1-2 libluajit-5.1-common
  libnetfilter-queue1 libpcrc3 net-tools oinkmaster snort-common
  snort-common-libraries snort-rules-default
Suggested packages:
  snort-doc
The following NEW packages will be installed:
  libdaq2t64 libdumbnet1 libluajit-5.1-2 libluajit-5.1-common
  libnetfilter-queue1 libpcrc3 net-tools oinkmaster snort snort-common
  snort-common-libraries snort-rules-default
0 upgraded, 12 newly installed, 0 to remove and 79 not upgraded.
Need to get 2,870 kB of archives.
After this operation, 12.2 MB of additional disk space will be used.
Get:1 http://my.archive.ubuntu.com/ubuntu noble/universe amd64 libluajit-5.1-common all 2.1.0+git20231223.c525bcb+dfsg-1 [49.2 kB]
Get:2 http://my.archive.ubuntu.com/ubuntu noble/universe amd64 libluajit-5.1-2 amd64 2.1.0+git20231223.c525bcb+dfsg-1 [275 kB]
Get:3 http://my.archive.ubuntu.com/ubuntu noble/universe amd64 libpcrc3 amd64 2:8.39-15build1 [248 kB]
Get:4 http://my.archive.ubuntu.com/ubuntu noble/universe amd64 snort-common-libraries amd64 2.9.20-0+deb11u1ubuntu1 [899 kB]
Get:5 http://my.archive.ubuntu.com/ubuntu noble/universe amd64 snort-rules-default all 2.9.20-0+deb11u1ubuntu1 [144 kB]
Get:6 http://my.archive.ubuntu.com/ubuntu noble/universe amd64 snort-common all 2.9.20-0+deb11u1ubuntu1 [47.7 kB]
```

3.3 Configuration of Network Variables

To ensure Snort correctly distinguishes between "internal" (safe) and "external" (hostile) traffic, the configuration file `/etc/snort/snort.conf` was modified.

Variables Adjusted:

- HOME_NET: Set to 192.168.158.0/24. This defines the internal network range that Snort is protecting.
- EXTERNAL_NET: Set to !\$HOME_NET. This logic instructs Snort to treat any IP address not belonging to the local subnet as potentially hostile.

Command used to edit:

```
sudo nano /etc/snort/snort.conf
```

```
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET 192.168.158.0/24

# Set up the external network addresses. Leave as "any" in most situations
ipvar EXTERNAL_NET !$HOME_NET
# If HOME_NET is defined as something other than "any", alternative, you can
# use this definition if you do not want to detect attacks from your internal
# IP addresses:
#ipvar EXTERNAL_NET !$HOME_NET
```

3.4 Custom Rule Development

While Snort comes with community rules, custom rules were written to meet the specific testing requirements of this project. These rules were added to `/etc/snort/rules/local.rules`.

Rule 1: ICMP Detection (Connectivity Check)

```
alert icmp any any -> $HOME_NET any (msg:"ICMP PING DETECTED - Possible Scan"; sid:1000001; rev:1; )
```

- **Rationale:** Detects ping sweeps used by attackers to check if a host is alive.

Rule 2: TCP Port Scan Detection (Stealth Scan)

```
alert tcp any any -> $HOME_NET any (flags: S; msg:"[!] NMAP PORT SCAN DETECTED [!]"; sid:1000002; rev:1;)
```

- **Rationale:** Detects TCP SYN packets (flags: S), which are characteristic of "Stealth Scans" (half-open connections) generated by tools like Nmap.

Rule 3: SSH Brute Force Detection

Bash

```
alert tcp any any -> $HOME_NET 22 (msg:"POTENTIAL SSH BRUTE FORCE ATTACK"; flags:S; flow:stateless; sid:1000004; rev:1;)
```

- **Rationale:** Monitors Port 22 for high volumes of connection attempts. This specific rule flags the rapid succession of login attempts characteristic of a `hydra` brute-force attack.

Rule 4: FTP Unauthorized Access Detection

Bash

```
alert tcp any any -> $HOME_NET 21 (msg:"FTP CONNECTION ATTEMPT DETECTED"; flags:S; sid:1000005; rev:1;)
```

- **Rationale:** Flags any attempt to initiate a connection to Port 21 (FTP). Since FTP transmits data in cleartext, unauthorized access attempts to this service are a critical security event.

Rule 5: Web Server Vulnerability Scanning

Bash

```
alert tcp any any -> $HOME_NET 80 (msg:"WEB SERVER ACCESS DETECTED"; flags:S; sid:1000006; rev:1;)
```

- **Rationale:** Monitors ingress traffic on Port 80 (HTTP). This allows the IDS to detect web-based reconnaissance, such as the vulnerability scanning performed by the `nikto` tool.

```
ubuntu@ubuntu-VMware-Virtual-Platform: ~/Desktop$ cat /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures.  Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:"ICMP PING DETECTED - Possible Scan"; sid:1000001; rev:1;)
alert tcp any any -> $HOME_NET any (flags: S; msg:"[!] NMAP PORT SCAN DETECTED [!]"; sid:1000002; rev:1;)
alert tcp any any -> $HOME_NET 22 (msg:"POTENTIAL SSH BRUTE FORCE ATTACK"; flags:S; flow:stateless; sid:1000004; rev:1;)
alert tcp any any -> $HOME_NET 21 (msg:"FTP CONNECTION ATTEMPT DETECTED"; flags:S; sid:1000005; rev:1;)
alert tcp any any -> $HOME_NET 80 (msg:"WEB SERVER ACCESS DETECTED"; flags:S; sid:1000006; rev:1;)
```

Section 4.0: Testing & Results

4.1 Test Methodology

The IDS was executed in "Console Mode" to provide real-time feedback during the demonstration. The following command was used to start the service:

```
sudo snort -A console -q -c /etc/snort/snort.conf -i ens33 -k none
```

4.2 Test Case A: ICMP Flooding

- Action: The attacker executed a standard connectivity check using ping 192.168.158.129.

```
(kali@kali)-[~/Desktop]
$ ping 192.168.158.129
PING 192.168.158.129 (192.168.158.129) 56(84) bytes of data.
64 bytes from 192.168.158.129: icmp_seq=1 ttl=64 time=0.503 ms
64 bytes from 192.168.158.129: icmp_seq=2 ttl=64 time=2.04 ms
64 bytes from 192.168.158.129: icmp_seq=3 ttl=64 time=2.26 ms
64 bytes from 192.168.158.129: icmp_seq=4 ttl=64 time=1.09 ms
64 bytes from 192.168.158.129: icmp_seq=5 ttl=64 time=1.56 ms
64 bytes from 192.168.158.129: icmp_seq=6 ttl=64 time=2.22 ms
64 bytes from 192.168.158.129: icmp_seq=7 ttl=64 time=1.10 ms
64 bytes from 192.168.158.129: icmp_seq=8 ttl=64 time=0.449 ms
64 bytes from 192.168.158.129: icmp_seq=9 ttl=64 time=0.875 ms
64 bytes from 192.168.158.129: icmp_seq=10 ttl=64 time=1.43 ms
64 bytes from 192.168.158.129: icmp_seq=11 ttl=64 time=1.65 ms
64 bytes from 192.168.158.129: icmp_seq=12 ttl=64 time=2.57 ms
64 bytes from 192.168.158.129: icmp_seq=13 ttl=64 time=1.99 ms
64 bytes from 192.168.158.129: icmp_seq=14 ttl=64 time=1.75 ms
64 bytes from 192.168.158.129: icmp_seq=15 ttl=64 time=0.640 ms
```

- Observation: Snort successfully triggered the custom rule sid:1000001.
- Log Output:

```
12/31-14:05:38.275793 [**] [1:1000001:1] ICMP PING DETECTED...
```



```

ubuntu@ubuntu-VMware-Virtual-Platform:~/Desktop$ sudo snort -A console -q -c /etc/snort/snort.conf -i ens33
12/31-14:05:38.275793 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.128 -> 192.168.158.129
12/31-14:05:38.275829 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.129 -> 192.168.158.128
12/31-14:05:39.290190 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.128 -> 192.168.158.129
12/31-14:05:39.290254 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.129 -> 192.168.158.128
12/31-14:05:40.292643 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.128 -> 192.168.158.129
12/31-14:05:40.292709 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.129 -> 192.168.158.128
12/31-14:05:41.294675 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.128 -> 192.168.158.129
12/31-14:05:41.294704 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.129 -> 192.168.158.128
12/31-14:05:42.296600 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.128 -> 192.168.158.129
12/31-14:05:42.296663 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.129 -> 192.168.158.128
12/31-14:05:43.299078 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.128 -> 192.168.158.129
12/31-14:05:43.299146 1:1000001:1 ICMP PING DETECTED - Possible Scan [**] [Priority: 0] {ICMP} 192.168.158.129 -> 192.168.158.128

```

4.3 Test Case B: Nmap Stealth Scan

- **Action:** The attacker executed a comprehensive port scan on the first 1,000 ports using the command: `sudo nmap -sS -p 1-1000 192.168.158.129`.

```

(kali@kali)-[~/Desktop]
$ sudo nmap -sS -p 1-1000 192.168.158.129
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-31 01:11 EST
Nmap scan report for 192.168.158.129
Host is up (0.00040s latency).
All 1000 scanned ports on 192.168.158.129 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 00:0C:29:98:4F:6D (VMware)

Nmap done: 1 IP address (1 host up) scanned in 0.24 seconds

```

Observation: Snort triggered the custom rule sid:1000002 immediately upon scan initiation. The console displayed a rapid stream of alerts corresponding to the scanned ports (e.g., 21, 22, 80, 443).

```

ubuntu@ubuntu-VMware-Virtual-Platform:~/Desktop$ sudo snort -A console -q -c /etc/snort/snort.conf -i ens33 -k none
12/31-14:11:58.649000 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:995
12/31-14:11:58.649005 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:25
12/31-14:11:58.649006 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:143
12/31-14:11:58.649007 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:21
12/31-14:11:58.649008 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:23
12/31-14:11:58.649009 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:443
12/31-14:11:58.649010 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:53
12/31-14:11:58.649010 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:80
12/31-14:11:58.649474 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:110
12/31-14:11:58.649476 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:22
12/31-14:11:58.650494 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:139
12/31-14:11:58.650497 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:135
12/31-14:11:58.650498 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:993
12/31-14:11:58.651114 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:113
12/31-14:11:58.651115 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:587
12/31-14:11:58.651116 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:111
12/31-14:11:58.651117 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:554
12/31-14:11:58.651117 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:199
12/31-14:11:58.651118 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:445
12/31-14:11:58.651119 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:256
12/31-14:11:58.651810 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:765
12/31-14:11:58.651812 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:201
12/31-14:11:58.651812 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:209
12/31-14:11:58.651813 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:669
12/31-14:11:58.651813 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:561
12/31-14:11:58.651814 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:398
12/31-14:11:58.651815 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:403
12/31-14:11:58.651815 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:656
12/31-14:11:58.652199 1:1000002:1 [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:213

```

4.4 Scenario D: SSH Brute Force Detection

Objective: To verify if the IDS can detect high-frequency authentication attempts

targeting the Secure Shell (SSH) service on Port 22.

Attacker Command (Kali):

```
hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.158.129 -t 4
```

```
(kali@kali)-[~/Desktop]
$ hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.158.129 -t 4
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations,
or for illegal purposes (this is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-01-04 03:42:45
[WARNING] Restorefile (you have 10 seconds to abort... (use option -I to skip waiting)) from a previous session found, t
o prevent overwriting, ./hydra.restore
[DATA] max 4 tasks per 1 server, overall 4 tasks, 14344399 login tries (l:1/p:14344399), ~3586100 tries per task
[DATA] attacking ssh://192.168.158.129:22/
```

Snort Detection Logs:

The following log snippet was captured during the attack execution:

```
ubuntu@ubuntu-Virtual-Platform:~/Desktop$ sudo snort -A console -q -u snort -g snort -c /etc/snort/snort.conf -i ens33
01/04-16:40:45.409199 ** [1:1000004:1] POTENTIAL SSH BRUTE FORCE ATTACK ** [Priority: 0] {TCP} 192.168.158.128:52222 -> 192.168.158.129:22
01/04-16:40:45.409199 ** [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:52222 -> 192.168.158.129:22
01/04-16:40:45.736677 ** [1:1000004:1] POTENTIAL SSH BRUTE FORCE ATTACK ** [Priority: 0] {TCP} 192.168.158.128:52228 -> 192.168.158.129:22
01/04-16:40:45.736677 ** [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:52228 -> 192.168.158.129:22
01/04-16:40:45.736677 ** [1:1000004:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:52228 -> 192.168.158.129:22
01/04-16:40:45.737683 ** [1:1000004:1] POTENTIAL SSH BRUTE FORCE ATTACK ** [Priority: 0] {TCP} 192.168.158.128:52226 -> 192.168.158.129:22
01/04-16:40:45.737683 ** [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:52226 -> 192.168.158.129:22
01/04-16:40:45.737684 ** [1:1000004:1] POTENTIAL SSH BRUTE FORCE ATTACK ** [Priority: 0] {TCP} 192.168.158.128:52236 -> 192.168.158.129:22
01/04-16:40:45.737684 ** [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:52236 -> 192.168.158.129:22
01/04-16:40:45.737866 ** [1:1000004:1] POTENTIAL SSH BRUTE FORCE ATTACK ** [Priority: 0] {TCP} 192.168.158.128:52250 -> 192.168.158.129:22
01/04-16:40:45.737866 ** [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:52250 -> 192.168.158.129:22
01/04-16:41:03.608686 ** [1:1000004:1] POTENTIAL SSH BRUTE FORCE ATTACK ** [Priority: 0] {TCP} 192.168.158.128:47454 -> 192.168.158.129:22
01/04-16:41:03.608686 ** [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:47454 -> 192.168.158.129:22
01/04-16:41:03.617153 ** [1:1000004:1] POTENTIAL SSH BRUTE FORCE ATTACK ** [Priority: 0] {TCP} 192.168.158.128:47458 -> 192.168.158.129:22
01/04-16:41:03.617153 ** [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:47458 -> 192.168.158.129:22
01/04-16:41:03.628942 ** [1:1000004:1] POTENTIAL SSH BRUTE FORCE ATTACK ** [Priority: 0] {TCP} 192.168.158.128:47460 -> 192.168.158.129:22
01/04-16:41:03.628942 ** [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:47460 -> 192.168.158.129:22
01/04-16:41:03.640477 ** [1:1000004:1] POTENTIAL SSH BRUTE FORCE ATTACK ** [Priority: 0] {TCP} 192.168.158.128:47468 -> 192.168.158.129:22
01/04-16:41:03.640477 ** [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] ** [Priority: 0] {TCP} 192.168.158.128:47468 -> 192.168.158.129:22
```

Analysis:

The IDS successfully identified the attack signature. The logs indicate a rapid succession of TCP connection attempts from the Attacker IP (**192.168.158.128**) targeting Port 22 on the Defender.

- **Double Trigger:** Interestingly, the traffic also triggered the **[!] NMAP PORT SCAN DETECTED [!]** rule. This confirms that the brute force tool (Hydra) generates traffic patterns similar to a TCP SYN scan, causing Snort to flag it as both a specific service attack and a general network anomaly.

4.5 Scenario E: FTP Service Access

Objective: To demonstrate the IDS's ability to detect unauthorized connection attempts to the File Transfer Protocol (FTP) service on Port 21.

Attacker Command (Kali):

```
ftp 192.168.158.129
```

```
(kali㉿kali)-[~/Desktop]
$ ftp 192.168.158.129
Connected to 192.168.158.129.
220 (vsFTPD 3.0.5)
Name (192.168.158.129:kali):
```

Snort Detection Logs:

The logs confirm that the connection initiated by the Attacker was immediately intercepted:

```
ubuntu@ubuntu-VMware-Virtual-Platform:~/Desktop$ sudo snort -A console -q -u snort -g snort -c /etc/snort/snort.conf -i ens33
01/04-17:00:24.734388  [**] [1:1000005:1] FTP CONNECTION ATTEMPT DETECTED [**] [Priority: 0] (TCP) 192.168.158.128:51386 -> 192.168.158.129:21
01/04-17:00:24.734388  [**] [1:1000002:1] [!] NMAP PORT SCAN DETECTED [!] [**] [Priority: 0] (TCP) 192.168.158.128:51386 -> 192.168.158.129:21
```

Analysis:

- The trigger of Rule `sid:1000005` confirms that Snort is correctly monitoring the vulnerable FTP port.
- The simultaneous trigger of Rule `sid:1000002` (Port Scan) highlights that the initial handshake of an FTP connection (TCP SYN) shares the same signature as a port scanner. This suggests that for a production environment, specific rules are needed to distinguish between legitimate user access and malicious scanning.

4.6 Scenario F: Web Server (HTTP) Vulnerability Scanning

Objective: To monitor and detect access attempts to the HTTP Web Server on Port 80. The test utilizes `nikto`, a comprehensive web server scanner, to simulate a hostile vulnerability assessment.

Attacker Command (Kali):

```
nikto -h 192.168.158.129
```

```
(kali㉿kali)-[~/Desktop]
$ nikto -h 192.168.158.129
- Nikto v2.5.0

+ Target IP: 192.168.158.129
+ Target Hostname: 192.168.158.129
+ Target Port: 80
+ Start Time: 2026-01-04 04:12:13 (GMT-5)

+ Server: Apache/2.4.58 (Ubuntu)
+ /: The anti-clickjacking X-Frame-Options header is not present. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Frame-Options
+ /: The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type. See: https://www.netsparker.com/web-vulnerability-scanner/vulnerabilities/missing-content-type-header/
+ No CGI Directories found (use '-C all' to force check all possible dirs)
+ /: Server may leak inodes via ETags, header found with file /, inode: 29af, size: 6478c3d033979, mtime: gzip. See: http://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2003-1418
+ OPTIONS: Allowed HTTP Methods: OPTIONS, HEAD, GET, POST
+ 8102 requests: 0 error(s) and 4 item(s) reported on remote host
+ End Time: 2026-01-04 04:12:38 (GMT-5) (25 seconds)

+ 1 host(s) tested
```

The scanner generated thousands of requests, which were successfully logged by the IDS.

```

ubuntu@ubuntu-VMware-Virtual-Platform:~/Desktop$ sudo snort -A console -q -u snort -g snort -c /etc/snort/snort.conf -i ens33
01/04-17:13:38.1398070100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:36570 -> 192.168.158.129:80
01/04-17:13:38.1398070100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:36570 -> 192.168.158.129:80
01/04-17:13:38.4261190100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:36582 -> 192.168.158.129:80
01/04-17:13:38.4261190100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:36582 -> 192.168.158.129:80
01/04-17:13:38.4772420100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:36590 -> 192.168.158.129:80
01/04-17:13:38.4772420100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:36590 -> 192.168.158.129:80
01/04-17:13:38.7187090100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:36600 -> 192.168.158.129:80
01/04-17:13:38.7187090100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:36600 -> 192.168.158.129:80
01/04-17:13:39.0236070100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:36610 -> 192.168.158.129:80
01/04-17:13:39.0236070100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:36610 -> 192.168.158.129:80
01/04-17:13:39.2749120100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:36624 -> 192.168.158.129:80
01/04-17:13:39.2749120100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:36624 -> 192.168.158.129:80
01/04-17:13:39.5517930100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41512 -> 192.168.158.129:80
01/04-17:13:39.5517930100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41512 -> 192.168.158.129:80
01/04-17:13:39.0019500100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41524 -> 192.168.158.129:80
01/04-17:13:39.0019500100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41524 -> 192.168.158.129:80
01/04-17:13:39.3557600100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41526 -> 192.168.158.129:80
01/04-17:13:39.3557600100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41526 -> 192.168.158.129:80
01/04-17:13:39.4195200100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41538 -> 192.168.158.129:80
01/04-17:13:39.4195200100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41538 -> 192.168.158.129:80
01/04-17:13:39.4293980100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41554 -> 192.168.158.129:80
01/04-17:13:39.4293980100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41554 -> 192.168.158.129:80
01/04-17:13:39.4343750100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41562 -> 192.168.158.129:80
01/04-17:13:39.4343750100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41562 -> 192.168.158.129:80
01/04-17:13:39.5478800100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41578 -> 192.168.158.129:80
01/04-17:13:39.5478800100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41578 -> 192.168.158.129:80
01/04-17:13:39.5515860100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41580 -> 192.168.158.129:80
01/04-17:13:39.5515860100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41580 -> 192.168.158.129:80
01/04-17:13:39.5554320100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41582 -> 192.168.158.129:80
01/04-17:13:39.5554320100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41582 -> 192.168.158.129:80
01/04-17:13:39.5651820100000000:1|WEB SERVER ACCESS DETECTED|**|Priority: 0|TCP|192.168.158.128:41596 -> 192.168.158.129:80
01/04-17:13:39.5651820100000002:1|!!|NMAPP ORT SCAN DETECTED|!!|**|Priority: 0|TCP|192.168.158.128:41596 -> 192.168.158.129:80

```

- **High-Volume Detection:** The logs show a rapid stream of alerts (multiple per second), confirming that `nikto` creates a distinct "noisy" traffic pattern.
- **Correlation:** Similar to the SSH and FTP attacks, the web traffic triggered both the service-specific rule (Rule 1000006) and the generic port scan rule (Rule 1000002). This demonstrates that the IDS is robust enough to handle high-load traffic without dropping packets during a scan.

The following log entry was captured during the attack and provides critical forensic information:

```
12/31-14:11:58.649010  [**] [1:1000002:1] [!!!] NMAP PORT SCAN DETECTED [!!!] [**] [Priority: 0] {TCP} 192.168.158.128:49022 -> 192.168.158.129:80
```

```
12/31-14:11:58.649000 [**] [1:1000002:1] [!!] NMAP PORT SCAN DETECTED [!!]  
[**] [Priority: 0] {TCP} 192.168.158.128:49022 → 192.168.158.129:80
```

- project

- Source IP (192.168.158.128): Identifies the attacker (Kali Linux).
 - Source Port (49022): An ephemeral port used by the attacker's Nmap tool to initiate the connection.
 - Destination IP (192.168.158.129): Confirms the target was our Ubuntu server.
 - Destination Port (80): Indicates the attacker was probing for a Web Server (HTTP).
 - Signature ID (1000002): Corresponds to our custom "Nmap Port Scan" rule.
-